

Graph Puzzles IV.2-preview: Four perfect matchings for snarks with a two-circuit 2-factor

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ABSTRACT: Let G be a snark distinct from the Petersen graph, and let M be a perfect matching such that $G - M$ consists of two circuits. We prove that M belongs to a cover of $E(G)$ by four perfect matchings. Consequently, G has perfect matching index four. The proof combines the Campos–Lucchesi theorem on separating cuts with the Karabáš–Máčajová theorem on Hamiltonian cubic 3-poles. We show that the cut between the two factor circuits is separating, use a perfect matching meeting this cut in three edges to reduce the graph, and extend a cover of the reduced graph along odd paths. The result applies, in particular, to every permutation snark distinct from the Petersen graph.

1 Introduction

The *perfect matching index* $\tau(G)$ of a bridgeless cubic graph G is the least number of perfect matchings whose union is $E(G)$. Three perfect matchings cover G precisely when they form the colour classes of a 3-edge-colouring. Covers by four perfect matchings are therefore the next case for snarks.

Graphs are finite and loopless. Throughout this paper, a *snark* is a connected, bridgeless, simple cubic graph of girth at least five and cyclic edge-connectivity at least four that has chromatic index four. A graph is cyclically 4-edge-connected if each edge-cut whose two shores contain circuits has size at least four. We write P_{10} for the Petersen graph. A *circuit* is a connected 2-regular subgraph, and a *2-factor* is a spanning 2-regular subgraph.

Karabáš and Máčajová [KM26]¹ proved that a cubic graph has a cover by four perfect matchings when it has a 2-factor consisting of two odd circuits joined by exactly three edges of the complementary perfect matching. Their Conjecture 3.9 asks whether every cubic graph with a two-circuit 2-factor has perfect matching index at most four, with the Petersen graph as the exception. We prove this assertion for snarks, with an additional conclusion concerning the complementary perfect matching.

¹ Ján Karabáš and Edita Máčajová. *On 4-perfect-matching covers of cubic graphs with two adjacent odd circuits in a 2-factor.* [arXiv:2605.09475](https://arxiv.org/abs/2605.09475) (2026).

Theorem 1.1. *Let $G \not\cong P_{10}$ be a snark, and let M be a perfect matching such that $G - M = A \cup B$ has exactly two circuit components. There are perfect matchings M_1, M_2, M_3 of G such that*

$$E(G) = M \cup M_1 \cup M_2 \cup M_3.$$

In particular, $\tau(G) = 4$.

An edge of M joining A and B is a *spoke*; one with both ends on one circuit is a *chord*. The theorem allows any number of chords. A snark with a 2-factor consisting of two induced circuits is called a *permutation snark*. We obtain the following consequence.

Corollary 1.2. *Every permutation snark distinct from P_{10} has perfect matching index four. For each 2-factor consisting of two induced circuits, its complementary perfect matching belongs to a cover by four perfect matchings.*

Proof. Apply theorem 1.1 to the complementary perfect matching of the given 2-factor. $\square$

The cut between the two factor circuits is separating. A theorem of Campos and Lucchesi gives a perfect matching meeting it in three edges. We suppress odd paths to obtain a graph covered by the three-spoke theorem, then extend its cover to G while retaining M .

2 Matching-covered contractions

For $X \subseteq V(G)$, the cut $\delta_G(X)$ consists of the edges with exactly one end in X ; the sets X and $V(G) \setminus X$ are its *shores*. A cut is *trivial* if one shore consists of a single vertex. We write G/X for the graph obtained by contracting X to one vertex and deleting the resulting loops. Contraction and suppression may produce parallel edges, which we retain as distinct edges. For an edge set S , let $V(S)$ be the set of its end vertices. Path length always counts edges.

We use the terminology of matching-covered graphs as in Lucchesi, de Carvalho, Kothari and Murty [LCKM18]². A graph is *matching covered* if it is connected and each edge belongs to a perfect matching. A cut is *separating* if both cut-contractions are matching covered. It is *tight* if every perfect matching meets it in exactly one edge. A *brick* is a nonbipartite matching-covered graph whose tight cuts are all trivial. A graph J is *factor-critical* if $J - v$ has a perfect matching for every $v \in V(J)$.

² C. L. Lucchesi, M. H. de Carvalho, N. Kothari and U. S. R. Murty. *On two unsolved problems concerning matching covered graphs*. SIAM Journal on Discrete Mathematics 32(2) (2018), 1478–1504. doi: 10.1137/17M1138704.

The following lemma will establish that the cut between A and B is separating.

Lemma 2.1. *Let H be a connected bridgeless loopless multigraph, and let $b \in V(H)$. Suppose that b has odd degree, every vertex of $H - b$ has degree three in H , and $H - b$ is factor-critical. Then H is matching covered.*

Proof. Each edge bu belongs to a perfect matching: take a perfect matching of $H - b - u$ and add bu . Consider an edge uv with $u, v \neq b$. Suppose that $H - u - v$ has no perfect matching. By Tutte's theorem [Tutte47]³, there is a set $X \subseteq V(H) \setminus \{u, v\}$ such that $H - X - u - v$ has more than $|X|$ odd components. Since $H - u - v$ has even order, the number q of these components satisfies

$$q \geq |X| + 2. \quad (1)$$

If $b \in X$, choose a perfect matching of $H - b - u$. Delete the $|X|$ vertices in $(X \setminus \{b\}) \cup \{v\}$ from this graph. Each odd component of the remaining graph contains a vertex whose partner in the chosen matching was deleted. Distinct odd components require distinct deleted partners. Hence $q \leq |X|$, contradicting (1).

If $b \notin X$, put $R = X \cup \{u, v\}$. All vertices of R have degree three, and $uv \in E(H[R])$, so

$$|\delta_H(R)| = 3|R| - 2|E(H[R])| \leq 3|X| + 4. \quad (2)$$

All vertices of H have odd degree. Each odd component K of $H - R$ therefore has an odd edge boundary. Connectedness and bridgelessness give $|\delta_H(V(K))| \geq 3$. These boundaries are disjoint subsets of $\delta_H(R)$, and (1) yields

$$|\delta_H(R)| \geq 3q \geq 3|X| + 6,$$

contrary to (2). Thus $H - u - v$ has a perfect matching. Adding uv proves that this edge also belongs to a perfect matching. $\square$

We also recall the standard connection between snarks and bricks. For cubic matching-covered graphs, every tight cut has size three; see Kothari, de Carvalho, Lucchesi and Little [KCLL20]⁴, Theorem 8 and Corollary 9. The argument below gives the needed consequence.

Lemma 2.2. *Every snark is a simple brick.*

Proof. Let G be a snark, and assign $x_e = 1/3$ to every $e \in E(G)$. This vector satisfies the degree equations $x(\delta_G(v)) = 1$ for all $v \in V(G)$. For every odd vertex set X , the cut $\delta_G(X)$ has odd size. Bridgelessness gives $|\delta_G(X)| \geq 3$, and hence $x(\delta_G(X)) \geq 1$. Edmonds' perfect matching polytope theorem [Edmonds65]⁵ therefore expresses x as a

³ W. T. Tutte. *The factorization of linear graphs*. Journal of the London Mathematical Society 22(2) (1947), 107–111. doi:10.1112/jlms/s1-22.2.107.

⁴ N. Kothari, M. H. de Carvalho, C. L. Lucchesi and C. H. C. Little. *On essentially 4-edge-connected cubic bricks*. Electronic Journal of Combinatorics 27(1) (2020), P1.22. doi:10.37236/8594.

⁵ Jack Edmonds. *Maximum matching and a polyhedron with 0,1-vertices*. Journal of Research of the National Bureau of Standards, Section B 69B (1965), 125–130. doi:10.6028/jres.069B.013.

convex combination of incidence vectors of perfect matchings. Since $x_e > 0$ for every edge, G is matching covered.

If D is a tight cut, every perfect matching has exactly one edge in D . The same convex combination gives

$$1 = x(D) = \frac{|D|}{3},$$

so $|D| = 3$. Cyclic 4-edge-connectivity implies that one shore of D induces a forest. If that forest has h vertices and c components, then its boundary has size

$$3h - 2(h - c) = h + 2c.$$

Since $h, c \geq 1$ and $h + 2c = 3$, the shore consists of a single vertex. Thus every tight cut is trivial. Finally, a cubic bipartite graph is 3-edge-colourable, so G is nonbipartite and hence a brick. $\square$

We use the following consequence of Campos and Lucchesi [CL00]⁶, Theorem 1.1. Their result is stated for near-bricks; the case of simple bricks is also recorded in [LCKM18], footnote 2 on p. 1496.

Theorem 2.3 (Campos–Lucchesi). *Let $G \not\cong P_{10}$ be a simple brick, and let C be a nontrivial separating cut of G . There is a perfect matching N of G such that*

$$|N \cap C| = 3.$$

We now apply lemma 2.1 to the given 2-factor.

Lemma 2.4. *Let G be a snark, and let M be a perfect matching for which $G - M = A \cup B$ has two circuit components. Then A and B are odd circuits, and $C = \delta_G(V(A))$ is a nontrivial separating cut.*

Proof. The circuits A and B have the same parity, because G has even order. If both were even, their edges could be coloured alternately with two colours and the edges of M with a third. Hence both are odd. All edges of C belong to M , so

$$|C| \equiv |V(A)| \equiv 1 \pmod{2}.$$

The cut separates two circuits. Cyclic 4-edge-connectivity therefore gives $|C| \geq 5$.

Consider $H = G/V(B)$, with contraction vertex b . Every vertex of $H - b$ has degree three in H , while $d_H(b) = |C|$ is odd. The graph $H - b$ contains the spanning odd circuit A , so deleting any vertex leaves a spanning path of even order and therefore a perfect matching. Thus $H - b$ is factor-critical. Each edge of A lies on that circuit, each chord lies on a circuit formed with a path of A , and each edge incident with

⁶ C. N. Campos and C. L. Lucchesi. *On the relation between the Petersen graph and the characteristic of separating cuts in matching covered graphs.* Technical Report IC-00-22, Institute of Computing, University of Campinas (2000). [report](#).

b lies on a circuit formed with another such edge and a path of A . Consequently, H is connected and bridgeless. By lemma 2.1, it is matching covered.

Interchanging A and B proves that $G/V(A)$ is matching covered as well. Thus C is separating. Both shores contain circuits of length at least five, so C is nontrivial. $\square$

3 The case of three spokes

A cubic 3-pole has three dangling edges, each with a single end vertex; every vertex is incident with three edges. A perfect matching of a 3-pole is a set of ordinary and dangling edges meeting each vertex exactly once. If a 3-pole H has a spanning circuit R , then $Q = E(H) \setminus E(R)$ is a perfect matching and contains all three dangling edges.

Theorem 3.1 of Karabáš and Máčajová [KM26] has the following form.

Theorem 3.1 (Karabáš–Máčajová). *Let H be a Hamiltonian cubic 3-pole with a distinguished spanning circuit R . There is a cover of H by four perfect matchings, one of which is $Q = E(H) \setminus E(R)$, such that every dangling edge belongs to exactly two of the four matchings.*

For the reduction in the next section, we need the conclusion with the complementary matching specified. It follows by gluing the two 3-poles, as in [KM26], Corollary 3.7.

Lemma 3.2. *Let H be a loopless cubic multigraph, and let S be a perfect matching such that $H - S$ consists of two odd circuits. If exactly three edges of S join the two circuits, there are perfect matchings L_1, L_2, L_3 of H such that*

$$E(H) = S \cup L_1 \cup L_2 \cup L_3.$$

Proof. Cut the three edges joining the circuits. Each circuit, together with its chords and three dangling edges, forms a Hamiltonian cubic 3-pole. Apply theorem 3.1 to each pole, using label 4 for the matching complementary to its spanning circuit and labels 1, 2, 3 for the other three matchings.

Each pole has odd order. A perfect matching of such a pole contains an odd number of dangling edges. In the cover supplied by theorem 3.1, the three dangling edges occur once each among the matchings labelled 1, 2, 3. Each of these three matchings therefore contains exactly one dangling edge. Permute the labels 1, 2, 3 on one pole so that the labels agree on the pairs of dangling edges that are to be joined. Gluing gives four perfect matchings of H . The matching labelled 4 is precisely S , and every edge is covered. $\square$

4 Proof of the main theorem

Theorem 1.1. *Let $G \not\cong P_{10}$ be a snark, and let M be a perfect matching such that $G - M = A \cup B$ has exactly two circuit components. There are perfect matchings M_1, M_2, M_3 of G such that*

$$E(G) = M \cup M_1 \cup M_2 \cup M_3.$$

In particular, $\tau(G) = 4$.

Proof. By lemma 2.2, G is a simple brick. By lemma 2.4, the cut $C = \delta_G(V(A))$ is nontrivial and separating. Theorem 2.3 supplies a perfect matching N with $|N \cap C| = 3$. Put

$$S = M \cap N.$$

Since $C \subseteq M$, exactly three edges of S join A and B . The edges of $N \setminus M$ form a perfect matching of

$$(A \cup B) - V(S).$$

Between consecutive vertices of $V(S)$ on either factor circuit, the internal vertices therefore occur in even number. Every path of the circuit between consecutive retained vertices has odd length.

Delete the edges of $M \setminus S$, and suppress the resulting vertices of degree two along A and B . Denote the resulting cubic multigraph by G' . Its retained matching is S , and $G' - S = A' \cup B'$ consists of two circuits. If c_A and c_B are the numbers of chords of S on A and B , respectively, then

$$|V(A')| = 3 + 2c_A, \quad |V(B')| = 3 + 2c_B.$$

Thus both circuits are odd and have at least three vertices. The graph G' is loopless, and exactly three edges of S join A' and B' . By lemma 3.2, it has a cover

$$E(G') = S \cup L_1 \cup L_2 \cup L_3$$

by four perfect matchings.

We extend this cover to G using the alternating assignment along odd paths from [KM26], Lemma 3.2. Give an edge e of G' the set of labels

$$\lambda(e) = \{i \in \{1, 2, 3\} : e \in L_i\} \cup \begin{cases} \{4\}, & e \in S, \\ \emptyset, & e \notin S. \end{cases}$$

At each vertex, the label sets on the three incident edges partition $\{1, 2, 3, 4\}$. Each set is nonempty. If $e \in E(A' \cup B')$, then $T = \lambda(e)$ is a subset of $\{1, 2, 3\}$ of size one or two, because the edge of S at either end carries label 4 and the other circuit edge has a nonempty label set.

Replace e by its original odd path in $A \cup B$. Give the successive edges of that path the label sets

$$T, \quad \{1, 2, 3\} \setminus T, \quad T, \quad \{1, 2, 3\} \setminus T, \quad \dots, \quad T. \quad (3)$$

Keep the labels on every edge of S , and give each restored edge of $M \setminus S$ the label set $\{4\}$. At a retained vertex, the incident label sets agree with those in G' . At a restored vertex, the two circuit edges carry complementary subsets of $\{1, 2, 3\}$, and the matching edge carries $\{4\}$. Hence each of the four labels occurs on exactly one edge incident with every vertex of G , and every edge carries at least one label.

For $i \in \{1, 2, 3\}$, let M_i be the set of edges carrying label i . These are perfect matchings of G , and the edges carrying label 4 are exactly the edges of M . Thus

$$E(G) = M \cup M_1 \cup M_2 \cup M_3.$$

A cover of a cubic graph by three perfect matchings assigns each of its three incident edges to a different matching at every vertex, and therefore gives a 3-edge-colouring. Since G is a snark, $\tau(G) = 4$. $\square$

References

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